

# MOHAMED HMINI

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Languages: English (fluent), French (professional), Arabic (native)

## INTRODUCTION

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Software Engineer with hands-on experience across Data Science, Data Engineering, DevOps, Machine Learning, and AI Engineering, holding degrees in Software Engineering and Intelligent Systems. Driven by problem solving and a preference for novel, unconventional challenges; my cross-disciplinary background enables effective collaboration across teams. Seeking a full-time AI/ML Engineering position.

## EXPERIENCE

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### 1. MLOps Engineer

Jan 2023 - Present

Viridien Group

Llandudno, UK

- Designed and built a proprietary event-driven highly scalable producer/consumer orchestration platform Docflow (internal) based on Celery, Redis, kvrocks, RabbitMQ and Keda; led the team's migration onto it through multiple stages — Spark (Nuclio) → Spark (offline inference) → Argo (Nuclio & offline) → Docflow (offline inference) — alongside a parallel migration from G2 (a proprietary HPC orchestration platform) and Airflow, I also leveraged the orchestrator's flexibility to support vLLM-style deployments, increasing throughput via scalable concurrent client calls.
- The platform is more than just an orchestration solution it is also a DevOps project with which I achieved a loosely-coupled software solution composed of many isolated and independent repositories (30 and growing), I also introduced a layered architecture that cleanly separates concerns across 7 layers: Gitlab repo, Python package, API layer, MCP layer, Docker image, Orchestrator wrapper, K8S wrapper (Helm), which freed developers from worrying about anything other than the code itself, it also enables Agnetic workflows to have access to our tooling.
- Built a Databricks-like self-service platform deploying per-user notebooks to K8s, letting developers spin up their own Spark clusters for distributed batch computing — complementing the orchestrator's event-driven approach for batch-driven workloads it doesn't fit.
- Built Sheet2Vec, a model-free, feature-driven embedding approach that extracts visual features from Excel XML to construct per-cell vectors, compressed via statistical methods into a single sheet-level vector — enabling clustering of visually similar sheets without semantic embeddings.
- Actively managing our deployments across client environments (Azure, Aws, on-prem), and maintaining our model registry using mlflow.
- (previous work) Built mdsdk (Model Deployment sdk), a drift-detection tool using alibi-detect deployed as a per-model sidecar alongside Nuclio model deployments, monitoring inference traffic in real time.
- (previous work) I built an end-to-end Model Deployment CI/CD template to be used to deploy all of our models to both Online and Offline modes.

### 2. Data Engineer

Apr 2022 - Jan 2023

Viridien Group

Llandudno, UK

- DevOps, coming up with adequate git branching strategies, helping with setting up CI/CD pipelines, dockerizing software.
- ML, modeling, building, testing and deploying machine learning models.
- Tools: Docker, Gitlab CI/CD, Python, Plotly Dash, Pytorch, etc

### 3. Data Engineer

Jan 2022 - Mar 2022

EVEAD Group

Casablanca, Morocco

- Building a google search based sitemap generator using selenium.
- Mimicking human behavior while browsing (dodging bot detections).

### 4. Machine Learning Intern

Mar 2021 - Sep 2021

LaBRI

Bordeaux, France

- Synthesis and learning of boolean networks for predicting cellular differentiation.
- Creating Different Boolean Networks' Embedding techniques to allow for a multilevel view of the same Boolean Networks' ensemble.
- Applying Feature Selection techniques to isolate the good Boolean Networks.
- Resynthesize new Boolean Networks using the learned rules.
- Using Machine Learning to Optimize a rule-based symbolic A.i system.
- Contributing to [BNeDiction](#) code base.

## EDUCATION

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**Master's Degree in Information Systems and Intelligent Systems (M2SI)**

INSEA, Morocco

9/2019 - 9/2021

**Bachelor's Degree in Information Systems and Big Data (BDSI)**

ESTS, Morocco

9/2018 - 6/2019

**DUT in Software Engineering**

ESTM, Morocco

10/2016 - 6/2018

## SAMPLE OF EARLY PERSONAL WORK

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**D-PandiSIM early 2021**

Big data based pandemics distributed simulator, scoring network nodes using inverse PageRank.

**Paysim Fraud Detection late 2020**

Fraud detection using VAE and BNN networks, generating distributions of the output.

**TweetsOLAPing early 2020**

Implementing an end-to-end tweets ETL/Analysis pipeline.

**IntelliWebWrapper early 2019**

Intelligent web content extraction, extracting the main List Region (list of products).